



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

29-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-I (ALL STAR BATCH)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	A	B	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C	A,B,C,D	A,D	A,B,D	A,D	A,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	6.00	10.00	0.00	124.00	7.00	0.75
SECTION-II (ii)	Q.	7	8	9			
	A.	3	3	5			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	D	C	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C	D	A,B,C,D	A,B,C	A,B,C,D	B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.20	0.89	196.00	286.00	5.00	4.00
SECTION-II (ii)	Q.	7	8	9			
	A.	4	6	4			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	C	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	C	B,D	A,B	A,B,C,D	B,D	A,B
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	6.00	10.00	4.00	6.00	2.00	9.00
SECTION-II (ii)	Q.	7	8	9			
	A.	1	2010	2022			

HINT – SHEET

PART-1 : PHYSICS

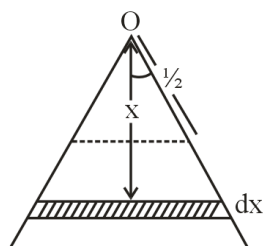
SECTION-I (i)

1. Ans (C)

$$dE = 2K \left(\frac{\sigma dx}{x} \right) \frac{1}{2}$$

$$E = \frac{\sigma}{4\pi\epsilon_0} \int_{\frac{\sqrt{3}}{4}}^{\frac{\sqrt{3}}{2}} \frac{dx}{x}$$

$$E = \frac{7}{44} \frac{\sigma}{\epsilon_0} \ln \sqrt{2}$$



2. Ans (A)

$$f = \frac{v}{2[\ell + 2 \times 0.6r]} = \frac{v}{2(\ell + 1.2r)}$$

$$\frac{\Delta f}{f} = \frac{\Delta v}{v} + \frac{\Delta(\ell + 1.2r)}{\ell + 1.2r}$$

$$= \frac{\Delta \ell + 1.2 \Delta r}{\ell + 1.2r}$$

$$\% \text{ error} = \frac{0.1 + 1.2(0.05)}{94 + 1.2 \times 5} \times 100 = 0.16 \%$$

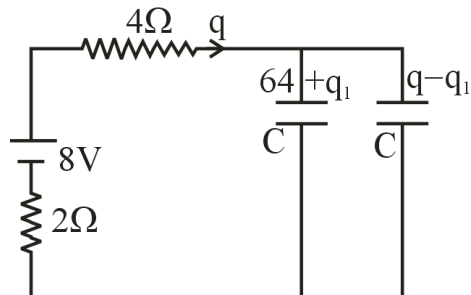
3. Ans (B)

By kirchoff's law.

$$8 = \frac{64 + q_1}{C} + 6i$$

$$\text{Also } 8 = \frac{q - q_1}{C} + 6i \quad i = \frac{dq}{dt}$$

$$q = \frac{3CE}{2} \left(1 - e^{-\frac{t}{(6 \times 2C)}} \right)$$



$$i = \frac{3E}{4(6)} e^{-\frac{t}{(6 \times 2C)}}$$

$$\text{Heat loss} = \int_0^{\infty} i^2 R dt = 384 \mu J$$

4. Ans (A)

$$\text{Initially } P_0 V = nRT_1 \dots (1)$$

$$\text{finally } \left(P_0 + \frac{W}{A} \right) (V - Ah) = nRT_2 \dots (2)$$

$$(Wh + P_0 h) = \Delta u = nC_v \Delta T \dots (3)$$

from equation (2)

$$2P_0 V - 2P_0 Ah = nRT_2$$

$$2nRT_1 - nRT_2 = 2P_0 Ah \dots (4)$$

from equation (3)

$$2P_0 Ah = n \left(\frac{3}{2} R \right) (T_2 - T_1) \dots (5)$$

$$\text{from equation (4)} \quad 2T_1 - T_2 = \frac{2P_0 Ah}{nR}$$

$$\text{from equation (5)} \quad T_2 - T_1 = \frac{4P_0 Ah}{3nR}$$

$$\text{Add above (2) equation } T_1 = \frac{P_0 Ah}{nR} \left(2 + \frac{4}{3} \right)$$

$$= \frac{10}{3} \frac{P_0 Ah}{nR} = 300K \quad T_2 = T_1 + \frac{4}{3} \frac{P_0 Ah}{nR}$$

$$= \frac{14}{3} \frac{P_0 Ah}{nR}$$

$$= 420 K$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,B,C)

Energy of incident photons are

$$E_1 = \frac{hc}{\lambda_1} = \frac{1240}{155} = 8 \text{ eV}$$

$$E_2 = \frac{1240}{310} = 4 \text{ eV}$$

$$E_3 = \frac{1240}{620} = 2 \text{ eV}$$

$$E_4 = \frac{1240}{1220} \simeq 1 \text{ eV}$$

work function = 1.4 eV

By Einstein equation

$$KE_{\max} = E_{\max} - \phi = 8 - 1.4$$

$$= 6.6 \text{ eV}$$

$$\text{Now applied p.d.} = \frac{E}{L} \cdot l$$

$$= \left(\frac{10}{100} \right) l$$

$$(A) KE = \left(6.6 - \frac{10}{100} \times 60 \right) \text{ eV} = 0.6 \text{ eV}$$

$$(B) KE = \left(6.6 - \frac{10}{100} \times 80 \right) \text{ eV} = \text{negative}$$

Which means no current

(C) For $l = 42 \text{ cm}$

$$KE_1 = 6.6 - \frac{10}{100} \times 42$$

$$= 2.4 \text{ eV}$$

for $l = 60 \text{ cm}$

$$KE_2 = 6.6 - \frac{10}{100} \times 60$$

$$= 0.6 \text{ eV}$$

$$KE_1 = 4 KE_2$$

(D) on changing terminals

$$KE = 6.6 + \frac{10}{100} \times L$$

$$KE_{\max} \text{ for } L = 100 \text{ cm}$$

$$= 16.6 \text{ eV}$$

6. Ans (A,B,C,D)

$$u = \frac{-K}{r^\alpha}$$

$$F = \frac{-du}{dr} = \frac{-K \propto}{r^{\alpha+1}}$$

$$\frac{mv^2}{r} = \frac{K \propto}{r^{\alpha+1}}$$

$$mv^2 - \frac{K \propto}{r^\alpha}$$

$$KE = \frac{K \propto}{2r^\alpha} = \frac{-\propto \mu}{2}$$

for Bohr's theory

$$mvr = \frac{nh}{2\pi}$$

$$v = \frac{nh}{2\pi mr}$$

from equation (1)

$$\frac{mn^2h^2}{4\pi^2m^2r^2} = \frac{K \propto}{r^\alpha}$$

$$r^{\alpha-2} = \frac{4\pi^2km \propto}{n^2h^2}$$

$$r = \left(\frac{n^2h^2}{4\pi^2Km \propto} \right)^{\frac{1}{2-\alpha}}$$

Total energy = KE + PE

$$= \frac{1}{2}mv^2 + PE$$

$$= \frac{1}{2}m \left(\frac{n^2h^2}{8\pi^2m^2r^2} \right) + \left(-\frac{K}{r^\alpha} \right)$$

$$= \frac{n^2h^2}{8\pi^2mr^2} - \frac{K}{r^\alpha}$$

7. Ans (A,D)

Ad equation

$$F_B = \rho_w v g = \frac{\rho_B v g}{3} = \frac{m_0 g}{3}$$

$$3\rho_w = \rho_B$$

$$m_0 g = T_0 + F_b$$

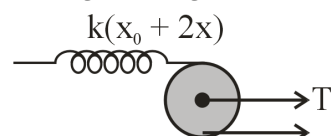
$$\frac{2}{3}m_0 g = T_0$$

$$T_0 = kx + f$$

$$kxR = fR$$

$$T_0 = 2kx$$

$$x = \frac{m_0 g}{3k} = \frac{2mg}{3k}$$



when slightly disp.

$$T - m_0 g - F_B = m_0 a$$

$$k(x_0 + 2x) + f - T = ma$$

$$k(x_0 + 2x)(2R) - TR = I_p a$$

$$a = \frac{4kx}{\left(\frac{I_p}{R^2} + m_0 \right)} = \frac{8kx}{7m}$$

8. Ans (A,B,D)

$$\ell_2 - \ell_1 = \frac{\lambda}{2} = 37.5 \text{ cm}$$

$$\lambda = 75 \text{ cm}$$

$$\Rightarrow \frac{\lambda}{4} = 18.75 \text{ cm}$$

$$e = 18.75 - 18 = 0.75 \text{ cm}$$

$$v = f\lambda = 375 \text{ m/s}$$

$$v = \sqrt{\frac{qRT}{M}}$$

$$q = 1.39$$

9. Ans (A,D)

$$\frac{L di}{dt} = vB\ell = \frac{dx}{dt} B\ell$$

$$L \int di = B\ell \int dx$$

$$Li = B\ell x$$

$$F = i\ell B = \left(\frac{B\ell x}{L} \right) \ell B$$

$$F \propto i\ell B$$

$$\text{Energy} = \frac{1}{2} Li^2 = \frac{1}{2} L \left(\frac{B\ell x}{L} \right)^2$$

$$\text{Energy} \propto x^2$$

10. Ans (A,D)

Voltage across capacitor

$$V = \varepsilon_0 \cos t \omega t \quad \omega t = \frac{t}{\sqrt{LC}} = \frac{10 \frac{\pi}{3} \times 10^{-3}}{\sqrt{10 \times 10 \times 10^{-6}}} = \frac{\pi}{3}$$

$$V = \varepsilon_0 \cos \frac{\pi}{3} = \frac{\varepsilon_0}{2} = 5 \text{ volt}$$

Since electric field at axis is zero (0), so, acceleration

also zero (0).

$$\text{By energy conservation } \frac{1}{2} C \varepsilon_0^2 = \frac{1}{2} C \left(\frac{\varepsilon_0}{2} \right)^2 + \frac{1}{2} Li^2$$

$$i = 8.7 \times 10^{-3} \text{ A}$$

$$B = \mu_0 ni$$

$$= 4\pi \times 10^{-7} \times 20,000 \times 8.7 \times 10^{-3}$$

$$= 218 \times 10^{-6}$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (6.00)

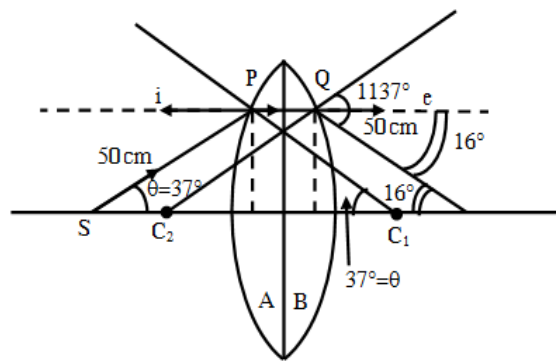
$$\frac{\lambda}{\sqrt{\frac{3mg}{m}}} = \frac{\lambda_0}{\sqrt{\frac{2mg}{m}}}$$

$$\lambda = \sqrt{\frac{3}{2}} \lambda_0 = 6 \text{ meter}$$

2. Ans (10.00)

$$v = \sqrt{\frac{(2m + \frac{m}{2})g}{\frac{m}{l}}} = \sqrt{\frac{5}{2}} gl = 10 \text{ m/s}$$

3. Ans (0.00)



By figure

$$i = \theta + \theta = 2\theta = 74^\circ$$

by shell's law

$$1 \sin 74 = \frac{8}{5} \sin \gamma$$

$$2 \left(\frac{3}{5} \right) \left(\frac{4}{5} \right) = \frac{8}{5} \sin \gamma$$

$$\gamma = 37^\circ$$

So light becomes parallel to principle axis
hence

$$D = \infty$$

$$\Rightarrow \frac{1}{D} = 0$$

4. Ans (124.00)

For next surface let angle of emergence is 'e'
and by symmetry angle of incidence = 37°

$$\text{Total deviation} = 53^\circ = (i - r_1) + (e - r_2)$$

$$r_1 = r_2 = 37^\circ \quad i = 74^\circ$$

$$53^\circ = 74^\circ - 37^\circ + e - 37^\circ$$

$$e = 53^\circ$$

Total length - SP + PQ + QI

$$SP = 50 \text{ cm}$$

$$PQ = 20 \text{ cm}$$

$$QI = \frac{30}{\sin 16^\circ}$$

$$\sin 16 = \sin(53 - 37)$$

$$= \frac{4}{5} \cdot \frac{4}{5} - \frac{3}{5} \cdot \frac{3}{5} = \frac{7}{25}$$

$$\text{Total length} = 50 + 20 + \frac{30}{\frac{7}{25}} \times 25$$

$$= 70 + \frac{750}{7} = \frac{490 + 750}{7}$$

$$\frac{10}{7} Q = \frac{1240}{7} \text{ cm}$$

$$Q = 124$$

5. **Ans (7.00)**

$$\begin{aligned}
 &= \left(\frac{1}{2} m R^2 \right) \omega_1 - \left(\frac{1}{2} m \left(\frac{R}{2} \right)^2 \right) \omega_2 \\
 &= \left[\frac{1}{2} m R^2 + \frac{1}{2} m \left(\frac{R}{2} \right)^2 + m \left(\frac{R}{2} \right)^2 \right] \omega_f \\
 &= \frac{w_1}{2} - \frac{w_2}{8}
 \end{aligned}$$

$$\begin{aligned}
 &= \left(\frac{1}{2} + \frac{1}{8} + \frac{1}{4} \right) \times \omega_f \\
 &= \frac{1}{2} - \frac{3}{8} = \frac{4+1+2}{8} \omega_f
 \end{aligned}$$

$$\beta = \omega_f = \frac{1}{7}$$

$$49\beta = 49 \times \frac{1}{7} = 7$$

6. **Ans (0.75)**

$$\Delta L = L_F - L_i$$

$$\begin{aligned}
 &= \left\{ \frac{1}{2} m \left(\frac{R}{2} \right)^2 \omega_2 + \frac{1}{8} \omega_1 \right\} \\
 &= \left\{ \frac{1}{2} m \left(\frac{R}{2} \right)^2 + m \left(\frac{R}{2} \right)^2 \right\} \omega_2 \\
 &\quad - \left\{ -\frac{1}{2} m \left(\frac{R}{2} \right)^2 \right\} \omega_1 \\
 &= m R^2 \left\{ \frac{3}{8} + \frac{3}{8} \right\} \\
 &= 10 \times 1 \times \frac{3}{4}
 \end{aligned}$$

$$\tau = \frac{\Delta L}{\Delta t} = 0.75$$

PART-1 : PHYSICS

SECTION-II (ii)

7. **Ans (3)**

Magnetic moment of Coil

$$M = NiA$$

$$= Ni (2R) (L)$$

$$\text{Net magnetic moment} = \sqrt{2}M$$

$$= \sqrt{2}Ni(2R)(L)$$

$$B = 1T$$

For Maximum Acceleration

$$\tau = \tau_{\max} \quad \theta = 90^\circ$$

$$\text{When } v = v_{\max} \quad \theta = 0^\circ$$

$$\Delta u = \Delta KE$$

$$(-MB \cos 90^\circ) - (-MB \cos 0)$$

$$= \frac{1}{2} MV^2 + \frac{1}{2} I \omega^2$$

$$+MB = \frac{1}{2} mv^2 + \frac{1}{2} \left(\frac{1}{2} m R^2 \right) \left(\frac{v^2}{R^2} \right)$$

$$\sqrt{2}Ni(2R)(L)B = \frac{3}{4} mv^2$$

$$\sqrt{2}(10)i \left(2 \times \frac{1}{\sqrt{2}} \right) (0.1)(1)$$

$$= \frac{3}{4} (1)(2)^2$$

$$i = \frac{3}{2} = 1.5A$$

8. **Ans (3)**

Fore due to water jet

$$F = \rho A (v - R\omega)^2 + \rho A (v - R\omega)^2$$

Now torque of find flow

$$\tau = \rho A \left\{ (v - R\omega)^2 + (v - R\omega)^2 \right\} R$$

Now Power

$$P = 2\rho A (v - R\omega)^2 R \cdot \omega$$

$$\frac{dP}{d\omega} = 0$$

$$\Rightarrow (v - R\omega)^2 - 2R(v - R\omega)\omega = 0$$

$$\Rightarrow (v - R\omega) \{ v - R\omega - 2R\omega \} = 0$$

$$\omega = \frac{v}{3R} = \frac{9}{3}$$

$$= 3 \text{ rads/s}$$

9. **Ans (5)**

Velocity of efflux will be $v = \sqrt{2gy}$ at any moment

$$\text{So, } \sqrt{2gy} a = -\pi \frac{y}{k} \frac{dy}{dt}$$

$$\int_{y_0}^0 \frac{\sqrt{y} dy}{\sqrt{2g}} = \int_0^t \frac{-ak}{\pi} dt$$

$$\frac{2}{3} y_0^{\frac{3}{2}} = \sqrt{2g} \frac{ak}{\pi} t$$

$$t = \frac{\pi}{3} \sqrt{\frac{2}{g}} \frac{y_0^{\frac{3}{2}}}{ak} = \frac{\pi}{3} \sqrt{\frac{2}{g}} \frac{(4)^{\frac{3}{2}}}{\pi \times \frac{1}{4}}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (B)**

In both cases, method-1 will yield $(\text{CH}_3)_4\text{N}^+\text{X}^-$ as major product.

In method-2 leaving group ability of $\text{TsO}^- > \text{I}^-$

2. **Ans (D)**

N_3H has H-bonding

3. **Ans (C)**

(A) $\text{MgO} > \text{CaO} > \text{SrO} > \text{BaO}$

(B) $\text{Na}_2\text{CO}_3 > \text{CaCO}_3 > \text{MgCO}_3 > \text{BeCO}_3$

(D) $\text{Fe}^{3+} > \text{Al}^{3+} > \text{Fe}^{2+}$

4. **Ans (B)**

$$M_1 = \frac{10}{11.35} ; M_2 = \frac{20}{11.35} ; M_3 = \frac{30}{11.35}$$

$$M = \frac{\frac{10}{11.35} \times 100 + \frac{20}{11.35} \times 100 + \frac{30}{11.35} \times 100}{1000}$$

$$= 0.53 \text{ M}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (B,C)**

Stable carbocation are formed in b & c. (a) is a hemiacetal.

6. **Ans (D)**

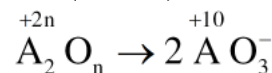
Only those metals which have lower B.P are refined by fractional distillation.

7. **Ans (A,B,C,D)**

For a -ve deviation solution

$$\Delta H < 0, \Delta S > 0, \Delta V < 0, \Delta G < 0$$

8. **Ans (A,B,C)**



Applying law of equivalence, $(10 - 2n) \times 1.5 \times 10^{-3}$

$$= 60 \times 0.03 \times 5 \times 10^{-3}$$

$$\Rightarrow n = 2 \text{ (a)}$$

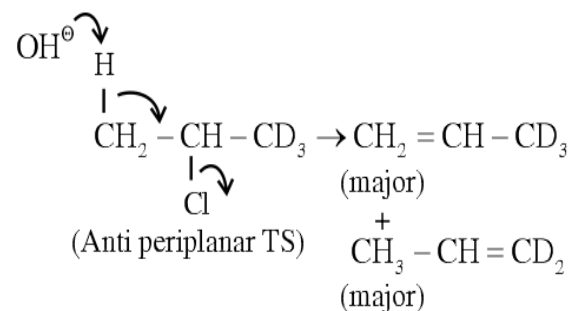
$$\Rightarrow \text{Empirical formula} = \text{AO}$$

$$\text{(c)} 0.15 \times 6 = x \times 6$$

$$\Rightarrow x = 0.15$$

(d) A may be some d-block element

9. **Ans (A,B,C,D)**



10. **Ans (B,C,D)**

$$\Delta p = \Delta x$$

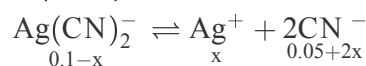
$$\Delta p \cdot \Delta x \geq \frac{h}{4\pi}$$

$$\Rightarrow \Delta x \geq \frac{1}{2} \sqrt{\frac{h}{\pi}}$$

PART-2 : CHEMISTRY

SECTION-II (i)

1. **Ans (1.20)**



$$3 \times 10^{-19} = \frac{x(0.05 + 2x)^2}{0.1 - x} \approx \frac{0.05^2 x}{0.1}$$

$$\Rightarrow x = 1.2 \times 10^{-17}$$

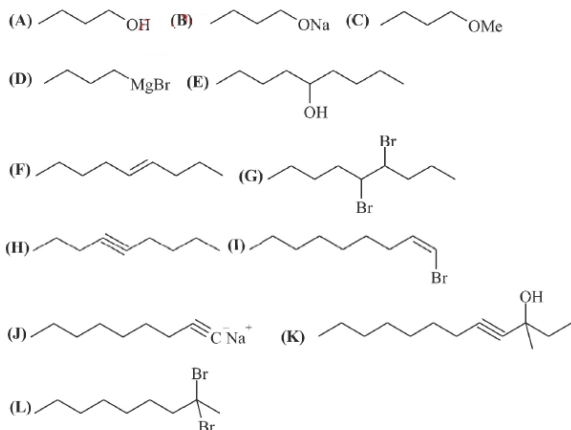
2. **Ans (0.89)**

$$E_{\text{Ag}^+/\text{Ag}} = E_{\text{Ag}^+/\text{Ag}}^0 + 0.1 \log 1.2 \times 10^{-17}$$

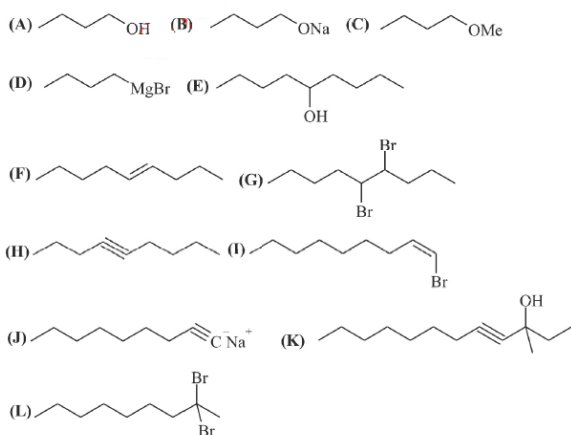
$$= 0.8 + 0.1 [0.6 + 0.48 - 18]$$

$$= -0.892 \text{ volts}$$

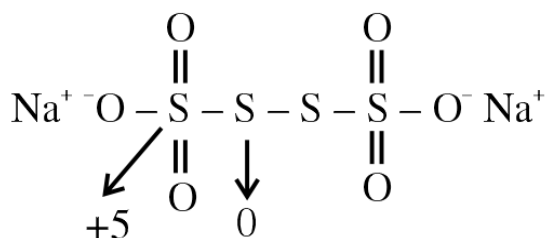
3. Ans (196.00)



4. Ans (286.00)



5. Ans (5.00)



6. Ans (4.00)

Phosphorus analogue $\Rightarrow \text{H}_4\text{P}_2\text{O}_6$

↓
 (2 types)
 Tetrabasic Tribasic
 Basicity = 4 Basicity = 3

PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (4)

a, b, d, e

8. Ans (6)

a, b, d, e, f, i

9. Ans (4)

$\sigma 1s, \sigma 3s, \pi^* 2p_y, \pi^* 2p_x$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$(21!)S$

$$\begin{aligned}
 &= {}^{21}C_1 - {}^{21}C_2 \left(1 + \frac{1}{2}\right) + {}^{21}C_3 \left(1 + \frac{1}{2} + \frac{1}{3}\right) - \dots - {}^{21}C_4 \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}\right) + \dots \\
 &= \sum_{r=1}^{21} (-1)^{r-1} {}^{21}C_r \left(1 + \frac{1}{2} + \dots + \frac{1}{r}\right) \\
 &= \sum_{r=1}^{21} (-1)^{r-1} {}^{21}C_r \int_0^1 (1+x+x^2+\dots+x^{r-1}) dx \\
 &= \int_0^1 \sum_{r=1}^{21} (-1)^{r-1} {}^{21}C_r (1+x+x^2+\dots+x^{r-1}) dx \\
 &= \int_0^1 \sum_{r=1}^{21} (-1)^{r-1} {}^{21}C_r \left(\frac{1-x^r}{1-x}\right) dx \\
 &= \int_0^1 \left(-\sum_{r=1}^{21} (-1)^r {}^{21}C_r + \sum_{r=1}^{21} (-1)^r {}^{21}C_r x^r\right) \times \frac{1}{1-x} dx \\
 &= \int_0^1 \left(-(-1) + (1-x)^{21} - 1\right) \frac{1}{1-x} dx \\
 &= \int_0^1 \frac{(1-x)^{21}}{1-x} dx \\
 &= \int_0^1 (1-x)^{20} dx \\
 &= -\frac{(1-x)^{21}}{21} \Bigg|_0^1 \\
 &= \frac{1}{21}
 \end{aligned}$$

2. Ans (C)

$$\begin{aligned}
 I &= \int_0^1 \frac{x^{a-1}}{1+x^b} dx \\
 &= \int_0^1 x^{a-1} (1+x^b)^{-1} dx \\
 &= \int_0^1 x^{a-1} (1 - x^b + x^{2b} - x^{3b} + x^{4b} - \dots) dx \\
 &= \int_0^1 (x^{a-1} - x^{a+b-1} + x^{2b+a-1} - \dots) dx \\
 &= \left(\frac{x^a}{a} - \frac{x^{a+b}}{a+b} + \frac{x^{2b+a}}{2b+a} - \dots \right)_0^1 \\
 &= \frac{1}{a} - \frac{1}{a+b} + \frac{1}{a+2b} - \frac{1}{a+3b} + \dots
 \end{aligned}$$

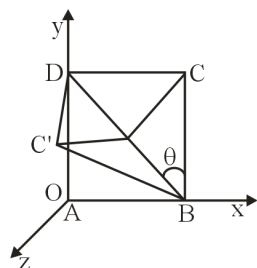
put $a = 1$ and $b = 3$ and integrate $\int_0^1 \frac{x^{a-1}}{1+x^b} dx$

3. Ans (C)

$$AB = CD = a$$

$$BC = AD = 2a$$

$$\tan \theta = \frac{CD}{BC} = \frac{1}{2}$$



N foot of perpendicular on BC drawn from C.

$$\sin \theta = \frac{CN}{BC} \Rightarrow \frac{1}{\sqrt{5}} = \frac{CN}{2a}$$

$$CN = C'N = \frac{2}{\sqrt{5}}a$$

$$\text{equation of BD in XY plane } \frac{x}{a} + \frac{y}{2a} = 1$$

and C is $(a, 2a)$

$$\text{coordinate of N are } \frac{x-a}{2} = \frac{y-2a}{1} = \frac{-(2a+2a-2a)}{5}$$

$$x = \frac{a}{5} \text{ and } y = \frac{8a}{5}$$

$$\text{Coordinate } C' \left(\frac{a}{5}, \frac{8a}{5}, \frac{2}{\sqrt{5}}a \right)$$

$$AC' = \frac{\sqrt{85}}{5}a$$

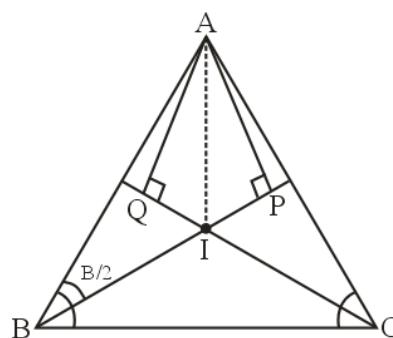
4. Ans (B)

In $\triangle APB$

$$\sin \frac{B}{2} = \frac{AP}{AB}$$

In $\triangle AQC$

$$\sin \frac{C}{2} = \frac{AQ}{AC}$$



Now in $\triangle ABI$

$$\frac{BI}{\sin \frac{A}{2}} = \frac{AB}{\sin \left(\pi - \frac{A}{2} - \frac{B}{2} \right)} = \frac{AB}{\cos \frac{C}{2}}$$

$$\text{In } \triangle ACI \text{ we have } \frac{CI}{\sin \frac{A}{2}} = \frac{AC}{\cos \frac{B}{2}}$$

$$\begin{aligned}
 \text{So, } \frac{AP}{BI} + \frac{AQ}{CI} &= \frac{\sin \frac{B}{2} \cdot \cos \frac{C}{2}}{\sin \frac{A}{2}} + \frac{\sin \frac{C}{2} \cdot \cos \frac{B}{2}}{\sin \frac{A}{2}} \\
 &= \frac{\sin \left(\frac{B+C}{2} \right)}{\sin \frac{A}{2}} = \cot \frac{A}{2}
 \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (C)

$$\int_0^x (x+1-t)f(t)dt = \int_0^x (4t^3 + 2t)dt$$

$$x \int_0^x f(t)dt + \int_0^x (1-t)f(t)dt = \int_0^x (4t^3 + 2t)dt$$

differentiate w.r.t x.

$$xf(x) + \int_0^x f(t)dt + (1-x)f(x) = 4x^3 + 2x$$

$$\int_0^x f(t)dt + f(x) = 4x^3 + 2x$$

Again differentiate

$$f(x) + f'(x) = 12x^2 + 2$$

$$\int_0^1 \frac{12}{f(x) + f'(x) + 10} dx$$

$$= \int_0^1 \frac{dx}{x^2 + 1}$$

$$= \tan^{-1}x \Big|_0^1 = \frac{\pi}{4}$$

6. Ans (B,D)

$$\vec{d} \cdot (\vec{a} + \vec{b} + \vec{c}) = 0, \quad \begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix} \neq 0$$

$$\Rightarrow \vec{d} \cdot \vec{a} + \vec{d} \cdot \vec{b} + \vec{d} \cdot \vec{c} = 0$$

Here,

$$\vec{d} = (\vec{a} \times \vec{b}) \sin x + (\vec{b} \times \vec{c}) \cos y + 2(\vec{c} \times \vec{a})$$

$$\vec{d} \cdot \vec{c} = \begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix} \sin x$$

$$\vec{d} \cdot \vec{a} = \begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix} \cos y$$

$$\vec{d} \cdot \vec{b} = \begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix} \cdot 2$$

adding

$$\vec{d} \cdot \vec{a} + \vec{d} \cdot \vec{b} + \vec{d} \cdot \vec{c} = \begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix} (\sin x + \cos y + 2)$$

$$\Rightarrow \sin x + \cos y + 2 = 0$$

$$\Rightarrow \sin x + \cos y = -2$$

$$\sin x = -1 \quad \cos y = -1$$

For minimum value of $x^2 + y^2$

$$x = -\frac{\pi}{2} \quad x = \pi$$

$$x^2 + y^2 = \frac{5\pi^2}{4}$$

$$\text{again } \frac{\vec{d} \cdot \vec{b}}{\begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix}} = 2$$

$$\Rightarrow \frac{\vec{d} \cdot (-\vec{a} - \vec{c})}{\begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix}} = 2$$

$$\Rightarrow \frac{\vec{d} \cdot (\vec{a} + \vec{c})}{\begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix}} = -2$$

7. Ans (A,B)

z_1, z_2 be the roots of equation $az^2 + bz + c = 0$

$$z_1 + z_2 = -\frac{b}{a}$$

$$z_1 z_2 = \frac{c}{a}$$

$$|z_1 z_2| = \left| \frac{c}{a} \right| = 1$$

$$|z_1 + z_2|^2 = \left| -\frac{b}{a} \right|^2 = 1$$

$$|z_1|^2 + |z_2|^2 + z_1 \bar{z}_2 + \bar{z}_1 z_2 = 1$$

$$z_1 \bar{z}_2 + \bar{z}_1 z_2 = -1$$

$$z_1^2 + z_2^2 = -z_1 z_2$$

$$(z_1 + z_2)^2 = z_1 z_2$$

$$\frac{b^2}{a^2} = \frac{c}{a} \Rightarrow b^2 = ac$$

$$\text{Again } \arg\left(\frac{z_1}{z_2}\right) = \theta \Rightarrow z_1 = z_2 e^{i\theta}$$

$$|z_1 + z_2| = |z_2 e^{i\theta} + z_2|$$

$$\Rightarrow 1 = |1 + e^{i\theta}|$$

$$\Rightarrow |1 + \cos \theta + i \sin \theta| = 1$$

$$(1 + \cos \theta)^2 + \sin^2 \theta = 1$$

$$2 + 2 \cos \theta = 1$$

$$\cos \theta = -\frac{1}{2}$$

$$\theta = \frac{2\pi}{3}$$

8. Ans (A,B,C,D)

$$\text{Here } f(x) - 2f\left(\frac{x}{2}\right) + f\left(\frac{x}{2^2}\right) = x^2$$

replace x by $\frac{x}{2}$

$$f\left(\frac{x}{2}\right) - 2f\left(\frac{x}{2^2}\right) + f\left(\frac{x}{2^3}\right) = \left(\frac{x}{2}\right)^2$$

Again replace x by $\frac{x}{2}$

$$f\left(\frac{x}{2^2}\right) - 2f\left(\frac{x}{2^3}\right) + f\left(\frac{x}{2^4}\right) = \left(\frac{x}{2^2}\right)^2$$

and so on

$$f\left(\frac{x}{2^n}\right) - 2f\left(\frac{x}{2^{n+1}}\right) + f\left(\frac{x}{2^{n+2}}\right) = \left(\frac{x}{2^n}\right)^2$$

adding

$$\begin{aligned} f(x) - f\left(\frac{x}{2}\right) - f\left(\frac{x}{2^{n+1}}\right) + f\left(\frac{x}{2^{n+2}}\right) \\ = x^2 + \frac{x^2}{4} + \frac{x^2}{4^2} + \dots + \frac{x^2}{4^n} \end{aligned}$$

as $n \rightarrow \infty$

$$f(x) - f\left(\frac{x}{2}\right) = x^2 \left(\frac{1}{1 - \frac{1}{4}} \right) = \frac{4x^2}{3} \dots (1)$$

$$f(x) - f\left(\frac{x}{2}\right) = \frac{4x^2}{3}$$

Again repeating same procedure in equation (1)

$$f(x) - f(0) = \frac{4}{3} x^2 \left(\frac{1}{1 - \frac{1}{4}} \right) = \frac{16x^2}{9}$$

$$f(x) = \frac{16}{9} x^2 + 1$$

9. Ans (B,D)

$$px^2 + 4\lambda xy + qy^2 + 4ax + 4ay + 4a = 0$$

represents a pair of straight line

$$\Rightarrow 4\lambda^2 - 4a\lambda + (p+q)a - pq = 0, \quad a \neq 0$$

$$\lambda \in \mathbb{R}, \quad 16a^2 - 4 \cdot 4((p+q)a - pq) \geq 0$$

$$a^2 - (p+q)a + pq \geq 0$$

$$(a-p)(a-q) \geq 0$$

$$a \leq p \quad \text{or} \quad a \geq q$$

10. Ans (A,B)

$$\left. \begin{array}{l} \text{Case (i)} \quad \left. \begin{array}{l} x < y < z \\ y < x < z \end{array} \right\} = 2^{n+1} C_3 \text{ ways} \end{array} \right\}$$

$$\text{Case (ii)} \quad x = y < z \Rightarrow {}^{n+1}C_2 \text{ ways}$$

$$\left. \begin{array}{l} \text{Case (iii)} \quad \left. \begin{array}{l} x < y = z \\ y < x = z \end{array} \right\} = 2^{n+1} C_2 \text{ ways} \end{array} \right\}$$

$$\text{Case (iv)} \quad x = y = z \Rightarrow n + 1 \text{ ways}$$

Total

$$\begin{aligned} & {}^{n+1}C_1 + 3 \cdot {}^{n+1}C_2 + 2 \cdot {}^{n+1}C_3 \\ &= {}^{n+1}C_1 + {}^{n+1}C_2 + 2({}^{n+1}C_2 + {}^{n+1}C_3) \\ &= {}^{n+2}C_2 + 2 \cdot {}^{n+2}C_3 \\ &= {}^{n+2}C_2 + {}^{n+2}C_3 + {}^{n+2}C_3 \\ &= {}^{n+3}C_3 + {}^{n+2}C_3 \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (6.00)

E_1 be the event that P_2 and P_7 are in same group

E_2 be the event that P_2 and P_7 are in different group.

$$P(E_1) = \frac{1}{11} \text{ and } P(E_2) = \frac{10}{11}$$

A be the event that exactly one of P_2 and P_7 among the losers.

$$P(A) = \frac{1}{11} \times 1 + \frac{10}{11} \left(\frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} \right) = \frac{6}{11}$$

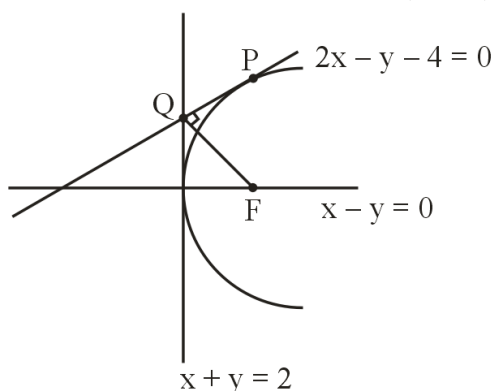
2. Ans (10.00)

Probability for both P_3 and P_5 are among

$$\text{winners} = \frac{10}{11} \times \frac{1}{2} \times \frac{1}{2} = \frac{5}{22}.$$

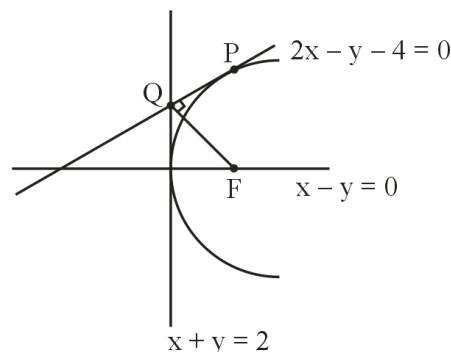
3. Ans (4.00)

In figure $PQ \perp QF$ co-ordinate of $f \left(\frac{2}{3}, \frac{2}{3} \right)$



4. Ans (6.00)

Mirror image of focus about any tangent is direction equation of directrix $3x + 3y - 8 = 0$



5. Ans (2.00)

$\alpha, \beta, \gamma, \delta$ roots of $x^4 - 5x^3 + kx^2 - 3x - 1 = 0$

$$(i) \quad (\alpha + \beta) + (\gamma + \delta) = 5$$

$$(ii) \quad (\alpha + \beta)(\gamma + \delta) + \alpha\beta + \gamma\delta = k$$

$$(iii) \quad \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) = 3$$

$$(iv) \quad (\alpha\beta)(\gamma\delta) = -1 \Rightarrow \alpha\beta = -1 \quad \gamma\delta = 1$$

$$\text{from (3)} \quad -(\gamma + \delta) + (\alpha + \beta) = 3$$

$$(\gamma + \delta) + (\alpha + \beta) = 5$$

$$\alpha + \beta = 4 \quad \gamma + \delta = 1$$

$$a = 4 \quad b = 1 \quad k = 4$$

Now $x^4 - 5x^3 + 4x^2 - 3x - 1 = (x^2 - 4x - 1)(x^2 - x + 1)$
equation has two real root and two imaginary roots
and $a + b + k = 9$

6. Ans (9.00)

$\alpha, \beta, \gamma, \delta$ roots of $x^4 - 5x^3 + kx^2 - 3x - 1 = 0$

$$(i) \quad (\alpha + \beta) + (\gamma + \delta) = 5$$

$$(ii) \quad (\alpha + \beta)(\gamma + \delta) + \alpha\beta + \gamma\delta = k$$

$$(iii) \quad \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) = 3$$

$$(iv) \quad (\alpha\beta)(\gamma\delta) = -1 \Rightarrow \alpha\beta = -1 \quad \gamma\delta = 1$$

$$\text{from (3)} \quad -(\gamma + \delta) + (\alpha + \beta) = 3$$

$$(\gamma + \delta) + (\alpha + \beta) = 5$$

$$\alpha + \beta = 4 \quad \gamma + \delta = 1$$

$$a = 4 \quad b = 1 \quad k = 4$$

Now $x^4 - 5x^3 + 4x^2 - 3x - 1 = (x^2 - 4x - 1)(x^2 - x + 1)$
equation has two real root and two imaginary roots
and $a + b + k = 9$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (1)**

Use weighted mean

$$\frac{\frac{1}{7}a^7 + \frac{2}{7}p^7 + \frac{4}{7}x^7}{\frac{1}{7} + \frac{2}{7} + \frac{4}{7}} \geq \left((a^7)^{\frac{1}{7}} (p^7)^{\frac{2}{7}} (x^7)^{\frac{4}{7}} \right)^{\frac{1}{\frac{1}{7} + \frac{2}{7} + \frac{4}{7}}}$$

$$\frac{1}{7}a^7 + \frac{2}{7}p^7 + \frac{4}{7}x^7 \geq ap^2x^4$$

similarly

$$\frac{1}{7}b^7 + \frac{2}{7}q^7 + \frac{4}{7}y^7 \geq bq^2y^4$$

$$\frac{1}{7}c^7 + \frac{2}{7}r^7 + \frac{4}{7}z^7 \geq cr^2z^4$$

$$\frac{1}{7}(a^7 + b^7 + c^7) + \frac{2}{7}(p^7 + q^7 + r^7) + \frac{4}{7}(x^7 + y^7 + z^7)$$

$$\geq ap^2x^4 + bq^2y^4 + cr^2z^4$$

$$1 \geq ap^2x^4 + bq^2y^4 + cr^2z^4$$

8. **Ans (2010)**

Here

$$\left. \begin{aligned} (x-2)a - b - c &= 0 \\ -a + (b-2)y - c &= 0 \\ -a - b + (z-2)c &= 0 \end{aligned} \right\} \text{homogenous equation}$$

has non trivial solution.

$$\begin{vmatrix} x-2 & -1 & -1 \\ -1 & y-2 & -1 \\ -1 & -1 & z-2 \end{vmatrix} = 0$$

$$(x-2)((y-2)(z-2)-1) + (-1)((z-2)-1) - 1(1+(y-2)) = 0$$

$$xyz - 2\Sigma xy + 3\Sigma x - 8 - \Sigma x + 6 - 2 = 0$$

$$xyz - 2\Sigma xy + 3\Sigma x - 4 = 0$$

$$xyz = 2\Sigma xy - 3\Sigma x + 4$$

$$= 2 \times 2011 - 3 \times 2012 + 4$$

$$= 4022 - 6036 + 4$$

$$= 2010$$

9. **Ans (2022)**

for $x > 0$ not possible

for $x < 0$ put $x = -y$

$$\left(1 - \frac{1}{y}\right)^{1-y} = \left(\frac{y-1}{y}\right)^{1-y}$$

$$= \left(\frac{y}{y-1}\right)^{y-1} = \left(\frac{2022}{2021}\right)^{2021}$$

$$\Rightarrow y = 2022$$

$$x = -2022$$

$$|x| = 2022$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

29-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : (SCORE-I)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,C	C,D	C,D	B,C	A,B,C	A,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	D	B	B	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	190.00	0.13	2250.00	23.40	9600.00	23.52
SECTION-II (ii)	Q.	7	8	9			
	A.	3	8	13			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,C,D	A,B,C	A,C	A,B,C,D	A,B	A,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	A	B	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	5.00	6.00 to 7.00	220.00	398.00	197.00	177.00
SECTION-II (ii)	Q.	7	8	9			
	A.	4	8	8			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C	A,B,D	A,D	A,C	A,B,C,D	A,B,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	C	A	C		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	2.00	4.00	7.00	71.00	623.00
SECTION-II (ii)	Q.	7	8	9			
	A.	108	7	309			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B,C)

$$R = P \frac{L}{A} \Rightarrow P = \frac{RA}{L}$$

$$P = R \frac{\pi r^2}{L} = \frac{5(\pi)(1 \times 10^{-3})^2}{1} = 5\pi \mu\Omega$$

Maximum relative error

$$\frac{\Delta p}{p} = \frac{\Delta R}{R} + \frac{2\Delta r}{r} + \frac{\Delta L}{L}$$

$$\frac{\Delta p}{p} = \frac{0.01}{5} + 2\left(\frac{0.01}{1}\right) + \frac{0.01}{1}$$

$$\Delta p = 5\pi(0.002 + 0.02 + 0.01) = 0.503$$

2. Ans (C,D)

$$h = \frac{2 S \cos \theta}{\rho r g} \Rightarrow h = 59.6 \text{ mm}$$

So it will fill the tube.

$$rh = Rh'$$

$$(0.25)(59.6) = R(25)$$

$$R = 0.596 \text{ mm} \approx 0.60 \text{ mm}$$

3. Ans (C,D)

Dielectric slab will move out from other side of capacitor plate so it is not periodic motion

Now capacitance of capacitor at 't'

$$C = \frac{\epsilon_0(a-x)a}{d} + \frac{k\epsilon_0 \times a}{d}$$

$$= \frac{\epsilon_0 a}{d} (kx + a - x)$$

Now energy of capacitor

$$u = \frac{1}{2} CV^2 = \frac{1}{2} \epsilon_0 a [(k-1)x + a] V^2$$

Slope of potential 'energy' Vs 'x'

$$\text{Slop} = \frac{1}{2} \epsilon_0 a V^2 (k-1)$$

Now for total energy

$$\text{force} = \frac{dy}{dx} = \frac{1}{2} \epsilon_0 a V^2 (k-1)$$

$$a = \frac{1}{2m} \epsilon_0 a v^2 (k-1)$$

$$v^2 - v_0^2 = 2ax \Rightarrow v^2 = v_0^2 + \frac{2\epsilon_0 a v^2}{2m} (k-1)x$$

$$T.E = \left[\frac{1}{2} m v_0^2 + \frac{1}{2} \epsilon_0 a v^2 (k-1)x \right] + \frac{1}{2} \epsilon_0 a [(k-1)x + a] v^2$$

4. Ans (B,C)

Force on the pine board F = rate of change of momentum

$$= \frac{10-1}{0.002} \text{ N}$$

$$= 4500 \text{ N}$$

Stress in the broad due to this force is

$$\sigma = \frac{F}{A} = \frac{4500}{9.0 \times 10^{-4}} \text{ N/m}^2$$

$$= 5.0 \times 10^6 \text{ N/m}^2$$

The board will break since stress of $5.0 \times 10^6 \text{ N/m}^2$ is more than the breaking stress of $3.6 \times 10^6 \text{ N/m}^2$.

5. Ans (A,B,C)

(A) At highest point and lowest point

acceleration is vertical

(B) Acceleration will be horizontal at one position.

(D) Acceleration at lowest point is directed towards O.

6. Ans (A,C)

$$i = i_0 (1 - e^{-\frac{t}{\tau}})$$

$$\tau = \frac{L}{R} = 0.3$$

$$t = 0.3 \ln 2$$

$$= i_0 (1 - e^{-\ln 2})$$

$$= \frac{i_0}{2} = \frac{\frac{1}{2} B \omega \ell^2}{2R} = \frac{B \omega \ell^2}{4R}$$

Torque on small element

$$d\tau = \int dF \cdot x$$

$$= \int_0^\ell (i dx B) x$$

$$= i \frac{\ell^2}{2} B$$

$$= \left(\frac{B \omega \ell^2}{4R} \right) \frac{\ell^2 B}{2}$$

$$= \frac{B^2 \ell^4 \omega}{8R} = \frac{(0.1)^2 (2)^4 \times 2}{8(0.01)}$$

$$\tau = 4 \text{ N} - \text{m}$$

$$\text{Similarly } t = 0.6 \ln 2 \quad i = i_0 \left(1 - \frac{1}{4} \right)$$

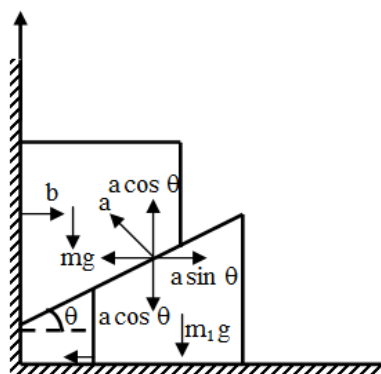
$$= \frac{3}{4} i_0 = \frac{3}{8} \frac{B \omega \ell^2}{R}$$

$$\tau = \frac{3}{16} \frac{B^2 \omega \ell^4}{R} = 6 \text{ N} - \text{m}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (D)



$$a \sin \theta = \mu(mg + a \cos \theta)$$

$$a(\sin \theta - \mu a \sin \theta) = \mu m_1 g$$

$$a \cos \theta + \mu a \sin \theta = m_2 g$$

$$\frac{\cos \theta + \mu \cos \theta}{\sin \theta - \mu \cos \theta} = \frac{m_2}{\mu m_1} = \frac{1}{\mu}$$

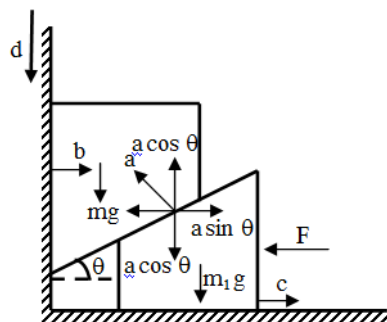
$$\frac{4}{5}\mu + \mu^2 \frac{3}{5} = \frac{3}{5} - \frac{4}{5}\mu$$

$$\frac{3}{5}\mu^2 + \frac{8}{5}\mu - \frac{3}{5} = 0$$

$$3\mu^2 + 8\mu - 3 = 0$$

$$\mu = \frac{-8 \pm \sqrt{64 + 36}}{6} = \frac{1}{3}$$

8. Ans (B)



$$F = \mu(mg + a \cos \theta) + a \sin \theta$$

$$a \cos \theta = \mu a \sin \theta + mg$$

$$a = \frac{mg}{\cos \theta - \mu \sin \theta} = \frac{mg}{\frac{4}{5} - \frac{1}{3} \cdot \frac{3}{5}} = \frac{5}{3}mg$$

$$\text{So } F = \frac{1}{3}(mg + \frac{5}{3}mg \cdot \frac{4}{5}) + \frac{5}{3}mg \cdot \frac{3}{5}$$

$$= \frac{7}{9}mg + mg = \frac{16}{9}mg$$

$$= 160 \text{ Newton}$$

9. Ans (B)

$$2g \rightarrow \frac{1}{6} \text{ mole} \rightarrow \frac{6.023 \times 10^{23}}{6 \times 10^{12}} \times 1.3$$

$$= 1.3 \times 10^{11} \text{ atoms}$$

$$\frac{dN}{dt} = \lambda N = \frac{\ln 2}{t_{\frac{1}{2}}} N$$

$$= \frac{\ln 2}{5733} \times 1.3 \times 10^{11} = \frac{\ln 2 \times 10^5}{2315}$$

10. Ans (D)

Activity when live ≈ 30 dpm.

Activity of prehistoric beam = 10 dpm

Activity decreasing $\frac{1}{3}$ times, which is possible

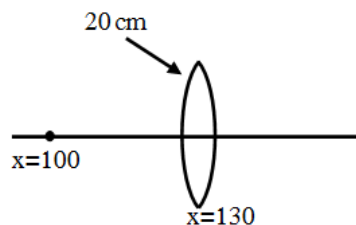
in between one and two half lives.

So 5000 - 10000 years

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (190.00)



$$\frac{1}{v} - \frac{1}{u} = (\mu - 1) \left(\frac{1}{R} - \frac{1}{-R} \right)$$

$$\frac{1}{v} = \frac{1}{-30} + \frac{1}{2} \times \frac{2}{+20} = \frac{1}{60}$$

$$x \text{ coordinate} = 130 + 60 = 190 \text{ cm}$$

2. Ans (0.13)

$$\frac{1}{v} - \frac{1}{u} = \frac{2(\mu - 1)}{R} \Rightarrow \frac{1}{v^2} \frac{dv}{dt} = \frac{1}{u^2} \frac{du}{dt} + \frac{2(\mu - 1)}{R^2} \frac{dR}{dt}$$

$$\frac{1}{(60)^2} \times (v - 12) = \frac{1}{(30)^2} \times (10 - 12) + \frac{2 \times \frac{1}{2}}{(20)^2} \times 1$$

$$v = 12 - 8 + 9 = 13 \text{ cm/s} = 0.13 \text{ m/s}$$

3. Ans (2250.00)

For BC process $P \propto V$ So, $P_B = 4P_0$

Now, $P_A V_A = P_B V_B$ & $P_A V_A^\gamma = P_C V_C^\gamma$ where

$$\gamma = 1.5$$

$$V_A^{\gamma-1} = \frac{P_C}{P_B} \times \frac{V_C^\gamma}{V_B}$$

$$\Rightarrow V_A^{1/2} = \frac{P_0}{2P_0} \times \frac{V_0^\gamma}{2V_0} \Rightarrow V_A = \frac{V_0}{16}$$

$$V_A = \frac{36000}{16} = 2250$$

4. Ans (23.40)

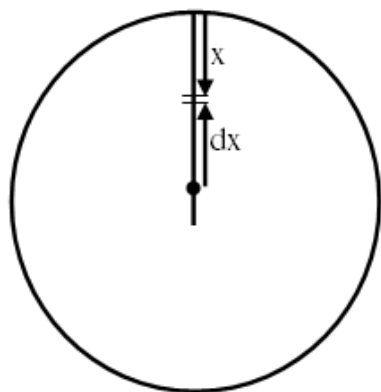
Work done = $W_{AB} + W_{BC} + W_{CA}$

$$= 4P_0 V_0 \ln \frac{2V_0}{\frac{V_0}{16}} + \frac{P_0 V_0 - 4P_0 V_0}{1+1} + \frac{4P_0 V_0 - P_0 V_0}{1-\frac{3}{2}}$$

$$= P_0 V_0 \left[20 \times \ln 2 - \frac{3}{2} - 6 \right]$$

$$= 10^5 \times 36 \times 10^{-3} \times \frac{13}{2} = 2340J$$

5. Ans (9600.00)



Centre of mass of wire will be at mid point. So centre of gravity has to be on surface

$$O = \frac{\int_0^l \mu dx g \left(1 - \frac{x}{R}\right) x}{\int_0^l \mu dx g \left(1 - \frac{x}{R}\right) x}$$

$$\frac{\ell^2}{2} - \frac{\ell^3}{3R} = 0 \Rightarrow \ell = \frac{3R}{2} = 9600km$$

6. Ans (23.52)

$$\text{Force} = \int_0^{3R/2} \mu dx g \left(1 - \frac{x}{R}\right) = \mu g \left[\frac{3R}{2} - \frac{9R}{8} \right] = \frac{3\mu g R}{8}$$

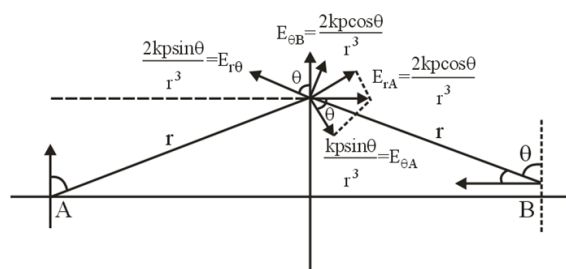
$$= \frac{3}{8} \times 10^{-3} \times 9.8 \times 6400$$

$$= 23.52$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (3)



$$\tan \theta = \frac{E_{rA}}{E_{\theta A}} = 2 \cot \theta \rightarrow \tan^2 \theta = 2 \rightarrow \tan \theta = \sqrt{2}$$

$$\text{Let } \frac{kp}{r^3} = E_0$$

$$E_y = E_0 \cos \theta \sin \theta + 2E_0 \sin \theta \cos \theta$$

$$= \frac{3}{4\pi\epsilon_0} \times \frac{q\ell \times \frac{\sqrt{2}}{\sqrt{3}} \times \frac{1}{\sqrt{3}}}{\frac{3\sqrt{3}}{2\sqrt{2}} \times d^3} = \frac{q\ell}{3\sqrt{3}\pi\epsilon_0 d^3}$$

8. Ans (8)

In steady state Rate of heat supply = Rate of heat lost via conduction + Rate of work done + Rate of increase of internal energy

Rate of heat loss at given instant

$$= \frac{T - 0}{t/K A} = \frac{(20 - 0)}{\frac{\sqrt{\pi}}{10}} \times 2\pi \times \frac{10}{\sqrt{\pi}} \times \frac{1}{2} \times 2 = 4000W$$

So,

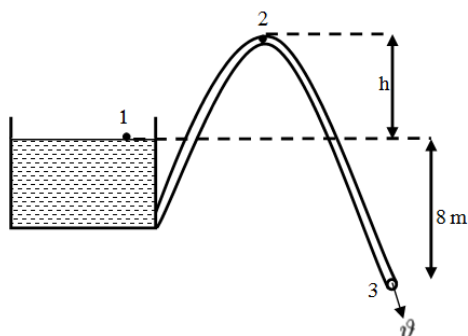
$$nC_p \frac{dT}{dt} = 4440 - 4000 = 440$$

$$\frac{5}{2} \frac{d}{dt} (nRT) = \frac{5}{2} P \cdot \frac{dV}{dt} = \frac{5}{2} (PA) v$$

$$\frac{5}{2} \frac{d}{dt} (nRT) = \frac{5}{2} P \cdot \frac{dV}{dt} = \frac{5}{2} (PA) v = \frac{5}{2} \times [P_{atm} A + mg] v = 440$$

$$v = 440 \times \frac{2}{5} \times \frac{1}{1100} = 0.16 \text{ m/s}$$

9. Ans (13)



Taking outlet level 3 as reference and applying Bernoulli's Equation between 1 and 3

$$P_{\text{atm}} + 0 + \rho g h = P_{\text{atm}} + \frac{1}{2} \rho v^2 + 0$$

$$\frac{v^2}{2g} = 8\text{m}$$

Applying Bernoulli's equation between 2 and 3

$$P_2 + \frac{1}{2} \rho v^2 + \rho g(h + 8) = P_{\text{atm}} + \frac{1}{2} \rho v^2 + 0$$

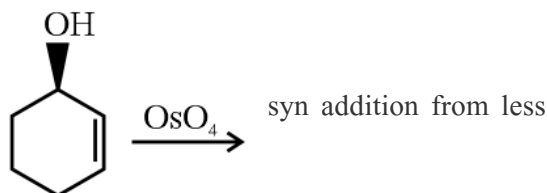
$$\frac{P_2}{\rho g} + h + 8 = \frac{P_{\text{atm}}}{\rho g} \rightarrow 1.0 + h + 8.0 = 10.3$$

$$h = 1.3\text{m}$$

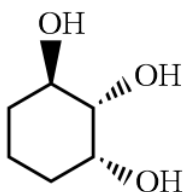
PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (B,C,D)



sterically hindered side gives



is formed.

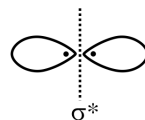
2. Ans (A,B,C)

For single e^- species, $1s < 2s = 2p < 3s = 3p = 3d$

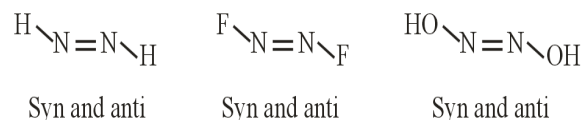
3. Ans (A,C)

Boron is diagonally related to Si

4. Ans (A,B,C,D)



5. Ans (A,B)



6. Ans (A,C)

Calcination involves heating ore in absence of air.

PART-2 : CHEMISTRY

SECTION-I (ii)

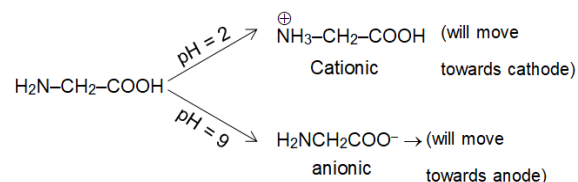
7. Ans (A)

$$pI = \frac{pK_{a1} + pK_{a2}}{2}$$

$$\text{Isoelectric point (PI)} = \frac{pK_{a1} + pK_{a2}}{2}$$

$$= \frac{1.9 + 3.7}{2} = \frac{5.6}{2} = 2.8$$

8. Ans (A)



9. Ans (B)

$$E_{\text{Sn}^{2+}/\text{Sn}} = -0.12 + \frac{0.06}{2} \log [\text{Sn}^{2+}]$$

$$= -0.12\text{V}$$

$$E_{\text{Ni}^{2+}/\text{Ni}} = -0.24 + \frac{0.06}{2} \log [\text{Ni}^{2+}]$$

$$= -0.24\text{V}$$

$$E_{\text{H}_2\text{O}/\text{H}_2} = -0.83 - \frac{0.06}{1} \log [\text{OH}^-]$$

$$[2\text{H}_2\text{O} + 2e^- \rightarrow \text{H}_2 + 2\text{OH}^-]$$

$$\Rightarrow E_{\text{H}_2\text{O}/\text{H}_2} = -0.83 + 0.42 = -0.41\text{V}$$

Hence, Sn^{2+} gets discharged.

10. **Ans (B)**

Under given conditions,

$$E_{\text{Sn}^{2+}/\text{Sn}} = E_{\text{Ni}^{2+}/\text{Ni}}$$

$$\Rightarrow -0.12 + \frac{0.06}{2} \log [\text{Sn}^{2+}] = -0.24$$

$$\Rightarrow \log [\text{Sn}^{2+}] = \frac{2 \times (-0.12)}{0.06} = -4$$

$$[\text{Sn}^{2+}] = 10^{-4} \text{ M}$$

PART-2 : CHEMISTRY

SECTION-II (i)

1. **Ans (5.00)**

5% of anion sites are vacant.

2. **Ans (6.00 to 7.00)**

$$\text{Density } (\rho) = \frac{4 \times 128}{6.023 \times 10^{23} \times (5)^3 \times 10^{-24}} = 6.80 \text{ g/cc}$$

3. **Ans (220.00)**

Initially, only NaOH was present at 0.1 M.

$$\begin{aligned} \Lambda_m(\text{NaOH}) &= \frac{k \times 1000}{C} \\ &= \frac{0.022 \times 1000}{0.1} \\ &= 220 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1} \end{aligned}$$

4. **Ans (398.00)**

After adding equal volume of HCl, NaOH will be neutralized completely giving NaCl of concentration 0.05 (since volume is doubled, concentration of salt will be halved).

$$\begin{aligned} \Lambda_{\text{NaCl}} &= \frac{0.0056 \times 1000}{0.05} \\ &= 112 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1} \end{aligned}$$

On further adding HCl, no neutralization will occur and conductivity will be due to NaCl and HCl together. Now the concentrations of NaCl = $\frac{2}{3} \times 0.05$ and concentration of

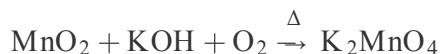
$$\text{HCl} = \frac{2}{3} \times 0.05 \text{ M.}$$

Now, $k = k(\text{NaCl}) + k(\text{HCl})$

$$\begin{aligned} &= (\Lambda_{\text{NaCl}} + \Lambda_{\text{HCl}}) \times \frac{C}{1000} \\ \Rightarrow \frac{1000 \times 0.017}{2 \times 0.05} \times 3 &= 112 + \Lambda_{\text{HCl}} \\ \Rightarrow \Lambda_{\text{HCl}} &= 510 - 112 = 398 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1} \end{aligned}$$

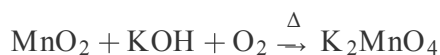
5. **Ans (197.00)**

P $\rightarrow$ NH₃, Q $\rightarrow$ NaPO₃, R $\rightarrow$ MnO₂, S $\rightarrow$ NaPCoO₄



6. **Ans (177.00)**

P $\rightarrow$ NH₃, Q $\rightarrow$ NaPO₃, R $\rightarrow$ MnO₂, S $\rightarrow$ NaPCoO₄

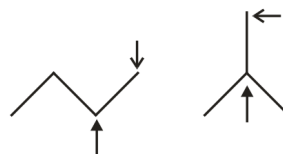


PART-2 : CHEMISTRY

SECTION-II (ii)

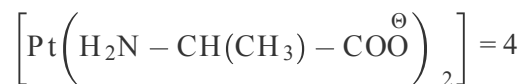
7. **Ans (4)**

$\Delta U = 0 \Rightarrow$ consider alcohols only



8. **Ans (8)**

Total number of geometrical isomers in



Total number of stereoisomer in $[\text{Pt}(\text{gly})_3]^\oplus = 4$
sum of geometrical isomers and stereoisomers of given complexes = 8.

9. **Ans (8)**

$$\Delta T_f = \frac{1000 K_f W_B}{M_B W_A}, M_B = 79X$$

$$79X = \frac{1000 \times 5.12 \times 2.47}{0.4 \times 50} = 632.32$$

$$X = \frac{632.32}{79} = 8$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A,B,C)**

Here none of z_i are zero and no two of z_i are equal i

$= 1, 2, 3, 4$

let $f(x) = (x - z_1)(x - z_2)(x - z_3)(x - z_4)$

$f(x) = (x - z_2)(x - z_3)(x - z_4) + (x - z_1)(x - z_3)(x - z_4)$

$+ (x - z_1)(x - z_2)(x - z_4) + (x - z_1)(x - z_2)(x - z_3)$

$f(z_1) = (z_1 - z_2)(z_1 - z_3)(z_1 - z_4) = \frac{4}{z_1}$

Similarly, $f(z_2) = \frac{4}{z_2}$

$f(z_3) = \frac{4}{z_3}, f(z_4) = \frac{4}{z_4}$

$\Rightarrow xf(x) = 4$ has roots z_1, z_2, z_3 & z_4

$\Rightarrow xf(x) - 4 = 0$ is a 4 degree polynomial has

leading coefficient is 4

$\Rightarrow xf(x) - 4 = 4f(x)$

$\Rightarrow \frac{f'(x)}{f(x) + 1} = \frac{4}{x}$

$\log(f(x) + 1) = 4 \log x + a$, a be any constant

$\log(f(x) + 1) = \log(ax^4)$

$f(x) + 1 = ax^4$

$f(x) = ax^4 - 1$ comparing coefficient of x^4 we get $a = 1$

$f(x) = x^4 - 1$

then the roots are 1, -1, i, -i

2. **Ans (A,B,D)**

$f(x) - mf(x) \leq 0 \quad \forall x \geq 1$

$\frac{d}{dx}(f(x)e^{-mx}) \leq 0$

Let $H(x) = f(x)e^{-mx}$

$H'(x) \leq 0 \quad \forall x \geq 1$

$x \geq 1 \quad H(x) \leq H(1)$,

$f(x)e^{-x} \leq 0$ but $f(x) \geq 0$

$f(x) \leq 0$

$\Rightarrow f(x) = 0 \quad x \geq 1$

3. **Ans (A,D)**

$f(x) = \tan x + \tan^2 x \cdot \frac{2 \tan x}{1 - \tan^2 x}$

$= \tan x \left(1 + \frac{2 \tan^2 x}{1 - \tan^2 x} \right)$

$= \tan x \left(\frac{1 + \tan^2 x}{1 - \tan^2 x} \right) = \frac{\tan x}{\cos 2x}$

$= \frac{\sin x}{\cos x \cdot \cos 2x} = \frac{\sin(2x - x)}{\cos x \cdot \cos 2x} = \tan 2x - \tan x$

$f(1) = \tan 2 - \tan 1$

$f(2) = \tan 4 - \tan 2$

$f(3) = \tan 6 - \tan 4$

$f(2^n) = \tan(2^{n+1}) - \tan(2^n)$

$g(n) = \tan 2^{n+1} - \tan 1$

4. **Ans (A,C)**

Here $P^T = -P$

$Q = (I + P)(I - P)^{-1}$

$Q^T = [(I + P)(I - P)^{-1}]^T$

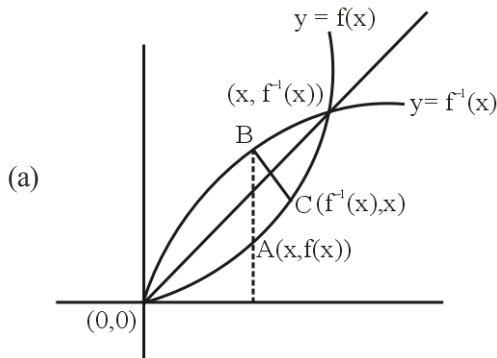
$= (I - P^T)^{-1} (I + P^T)$

$= (I + P)^{-1} (I - P)$

$|Q||Q^T| = 1, |Q| = 1, |2Q| = 2^3|Q| = 8$

$|\text{adj} Q| = |Q|^{3-1} = 1$

5. Ans (A,B,C,D)



Slope of OC > slope OA

$$\frac{x}{f^{-1}(x)} > \frac{f(x)}{x} \Rightarrow x^2 > f^{-1}(x) f(x) \quad \forall x \in (0,1)$$

(b) Here $f(a) = f(\beta)$

$$\Rightarrow f'(x) = 0 \text{ has at least one real } \gamma \in (a, \beta)$$

$f(x) \cdot f'(x) = 0$ has roots a, γ, β

and its derivative $(f'(x))^2 + f(x)f''(x) = 0$ has at least two real root in (a, β) .

(c) Here $g(a) = g(b) = 0$

let $x = c$ is a point of global maxima $C \in (a, b)$

$$\text{then } g'(c) = 0 \quad g''(c) < 0$$

$$\text{but } g''(c) + g'(c) \cdot g(c) = g(c)$$

$$g''(c) = g(c) < 0$$

$$\text{Since } g(a) = g(b) = 0$$

$$\Rightarrow \text{maximum value of } g(x) = 0 \quad \forall x \in (a, b)$$

(d) Here consider $g(x) = f^2(x) f(1-3x)(1-2x)$

$$\text{apply R.T. on } \left[\frac{1}{3}, \frac{1}{2} \right]$$

6. Ans (A,B,C,D)

$$(a) f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + x \log \left(\frac{1-x+x^2}{1+x+x^2} \right)$$

$$f(-x) = \frac{-xe^x}{1-e^x} - \frac{x}{2} + x \log \left(\frac{1-x+x^2}{1+x+x^2} \right)$$

$$f(-x) = \frac{xe^x - x + x}{e^x - 1} - \frac{x}{2} + x \log \left(\frac{1-x+x^2}{1+x+x^2} \right)$$

$$f(-x) = f(x) \text{ is even}$$

(b) $f(x)$ is invertible $\Rightarrow f(x)$ must be one one

$$f'(x) = x^2 + x + a > 0$$

$$\text{Disc} < 0$$

$$1 - 4a < 0$$

$$a \geq \frac{1}{4}$$

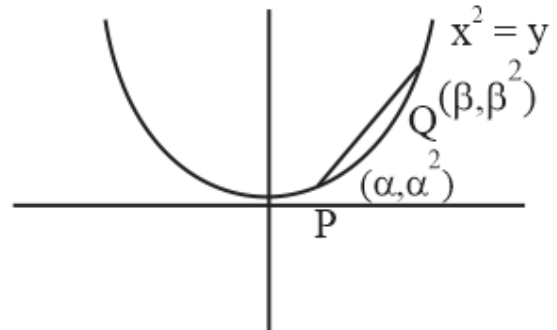
(c) Use $f(x) = f^{-1}(x)$ lies on line $y = x$

(d) $z^n = \bar{z}$ has $n+2$ solution.

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B)



$$\text{Equation of PQ, } y - \alpha^2 = (\alpha + \beta)(x - \alpha)$$

$$y = (\alpha + \beta)x - \alpha\beta$$

$$\text{Required area } \int_{\alpha}^{\beta} ((\alpha + \beta)x - \alpha\beta) - x^2 dx = \frac{4}{3}$$

$$\Rightarrow \int_{\alpha}^{\beta} (x^2 - x(\alpha + \beta))$$

$$+ \frac{(\alpha + \beta)^2}{4} - \frac{(\alpha + \beta)^2}{4} + \alpha\beta) dx = -\frac{4}{3}$$

$$\int_{\alpha}^{\beta} \left(x - \frac{\alpha + \beta}{2} \right)^2 dx$$

$$- \frac{(\alpha - \beta)^2}{4} \int_{\alpha}^{\beta} dx = -\frac{4}{3}$$

$$\left[\frac{\left(x - \frac{\alpha + \beta}{2} \right)^3}{3} \right]_{\alpha}^{\beta} - \frac{(\beta - \alpha)^3}{4} = -\frac{4}{3}$$

$$\frac{(\beta - \alpha)^3}{3 \times 8} - \frac{(\alpha - \beta)^3}{3 \times 8} - \frac{(\beta - \alpha)^3}{4} = -\frac{4}{3}$$

$$\left(\frac{1}{12} - \frac{1}{4} \right) (\beta - \alpha) = -\frac{4}{3}$$

$$-\frac{1}{6} (\beta - \alpha)^3 = \frac{-4}{3}$$

$$\beta - \alpha = 2$$

8. **Ans (C)**

Equation of PQ $y = (\alpha + \beta)x - \alpha\beta$

Let middle point of PQ is (h, k)

$$\text{Equation of PQ } xh - \left(\frac{y+k}{2}\right) = h^2 - k$$

$$xh - \frac{y}{2} = h^2 - \frac{k}{2}$$

$$\frac{y}{2} = hx - h^2 + \frac{k}{2}$$

$$y = 2hx + k - 2h^2$$

Comparing with $y = (\alpha + \beta)x - \alpha\beta$

$$\alpha + \beta = 2h \quad \alpha\beta = 2h^2 - k$$

$$\beta - \alpha = 2 \quad h^2 - 1 = 2h^2 - k$$

$$\alpha = h + 1 \quad h^2 = k - 1$$

$$\beta = h - 1 \quad x^2 = y - 1$$

9. **Ans (A)**

$$f(x) = \int_0^1 e^{|t-x|} dt, \quad 0 \leq x \leq 1$$

$$= \int_0^x e^{x-t} dt + \int_x^1 e^{t-x} dt$$

$$= -e^{x-t} \Big|_0^x + e^{t-x} \Big|_x^1$$

$$= -1 + e^x + e^{1-x} - 1$$

$$f(x) = e^x + e^{1-x} - 2$$

$$f'(x) = e^x - e^{1-x} = 0 \Rightarrow x = \frac{1}{2}$$

f decreases in $\left[0, \frac{1}{2}\right]$

f increases in $\left(\frac{1}{2}, 1\right]$

$$\text{Now } f(0) = e - 1$$

$$f(1) = e - 1$$

Maximum of $f(x) = e - 1$

10. **Ans (C)**

From the above solution

$$f(x) \text{ has minima at } x = \frac{1}{2}$$

$$\Rightarrow f\left(\frac{1}{2}\right) = 2(\sqrt{e} - 1)$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (2.00)**

$$(1 + f(x)) f''(x) = 1 + x \text{ for } x > -1$$

$$(1 + f(x)) f'(x) > 0 \quad \forall x > -1$$

$$1 + f(x) < 0 \text{ \& } f''(x) < 0$$

$$\text{or } 1 + f(x) > 0 \text{ \& } f''(x) > 0 \quad \forall x > -1$$

$$\text{but } f(0) = 1 \text{ so } 1 + f(x) < 0$$

$$\Rightarrow 1 + f(x) > 0 \text{ \& } f''(x) > 0$$

$f'(x)$ is increasing

$$\Rightarrow f'(x) > f'(0) \quad \forall x > 0$$

$$f'(x) > 0 \quad \forall x > 0$$

$$\Rightarrow f(x) \text{ is increasing in } (0, \infty)$$

$$f(x) \geq 1 \quad \forall x \geq 0$$

$$\text{Now, } f''(x) = \frac{1+x}{1+f(x)} \leq \frac{1+x}{2}$$

$$\int_0^x f''(x) dx \leq \int_0^x \frac{1+x}{2} dx$$

$$f'(x) \leq \frac{x}{2} + \frac{x^2}{4} \Rightarrow f'(2) \leq 2$$

2. **Ans (2.00)**

$$\int_0^x f'(x) dx \leq \int_0^x \left(\frac{x}{2} + \frac{x^2}{4}\right) dx$$

$$\leq \left[\frac{x^2}{4} + \frac{x^3}{12}\right]_0^x$$

$$\Rightarrow f(x) \leq \frac{x^2}{4} + \frac{x^3}{12}$$

$$f(2) \leq \frac{32}{12}$$

3. Ans (4.00)

$$\frac{d}{dx}(g^{-1} \circ f^{-1}(x)) = g^{-1}'(f^{-1}(x)) \cdot f^{-1}'(x) \text{ here } f^{-1}(-1) = -3$$

$$\frac{d}{dx}(g^{-1} \circ f^{-1}(x))_{x=-3} = g^{-1}'(-1) \cdot f^{-1}'(-3) \dots (I)$$

$$\text{Again } f^{-1}(f(x)) = x$$

$$f^{-1}(f(x)) \cdot f'(x) = 1 \Rightarrow f^{-1}'(f(x)) = \frac{1}{4 + \pi \sin \pi x}$$

$$\text{at } x = -1, f^{-1}'(-3) = \frac{1}{4} \dots (II)$$

$$f^{-1}(-1) = \frac{1}{4}$$

$$f(0) = -1$$

$$g^{-1}(g(x)) = x \quad \text{but } g(0) = -1$$

$$g^{-1}'(g(x)) \cdot g'(x) = 1 \quad g^{-1}'(-1) = 0$$

$$g^{-1}'(g(x)) = \frac{1}{3x^2 + 4 + \pi \sin \pi x} \Rightarrow g^{-1}'(g(-1)) = \frac{1}{7}$$

$$\text{at } x = 0 \quad g^{-1}'(g(0)) = \frac{1}{4} \Rightarrow g^{-1}'(-4) = \frac{1}{7}$$

$$g^{-1}'(-1) = \frac{1}{4} \dots (III)$$

Using equation (II), (III) in (I)

$$\text{We get } \frac{d}{dx}(g^{-1} \circ f^{-1}(x))_{x=-3} = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

4. Ans (7.00)

$$\text{We get } \frac{d}{dx}(g^{-1} \circ f^{-1}(x))_{x=-3} = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

$$\frac{d}{dx}(f^{-1} \circ g^{-1}(x)) = f^{-1}'(g^{-1}(x)) \cdot g^{-1}'(x)$$

$$\frac{d}{dx}(f^{-1} \circ g^{-1}(x))_{\text{at } x=-4} = f^{-1}'(g^{-1}(-4)) \cdot g^{-1}'(-4)$$

$$\text{Here } g(-1) = -4 \quad g^{-1}(-4) = -1$$

$$\frac{d}{dx}(f^{-1} \circ g^{-1}(x))_{\text{at } x=-4} = f^{-1}'(-1) \cdot g^{-1}'(-4)$$

$$= \frac{1}{4} \times \frac{1}{7} = \frac{1}{28}$$

5. Ans (71.00)

$$\text{Total case} = 2^5 + 5 \times 2^4$$

$$\text{probability} = \frac{2^5 + 5 \times 2^4}{2^{10}} = \frac{7}{2^6} = \frac{7}{64}$$

6. Ans (623.00)

Probability for at least four consecutive tall

$$= \frac{2^5 + 5 \times 2^4 - 1}{2^9} = \frac{111}{512}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (108)

$$\text{Let } \vec{b} = x\vec{a} + y\vec{c} + z\vec{d} \dots (i)$$

$$\text{dot with } \vec{b}, 1 = \frac{-x}{11} - \frac{y}{11} - \frac{z}{11}$$

$$\Rightarrow x + y + z = -11 \dots (ii)$$

$$\text{dot with } \vec{c}, -\frac{1}{11} = -\frac{x}{11} + y - \frac{z}{11}$$

$$x + z = 1 + 11y \dots (iii)$$

$$\text{from equation (ii) and (iii) } y = -1$$

$$\text{From equation (i) } \vec{b} + \vec{c} = x\vec{a} + z\vec{d} \dots (iv)$$

$$\text{dot with } \vec{a} \text{ in equation (iv),}$$

$$-\frac{1}{11} - \frac{1}{11} = x + z(\vec{d} \cdot \vec{a})$$

$$\Rightarrow \frac{-2}{11} = x + z(\vec{d} \cdot \vec{a})$$

$$\text{dot with } \vec{d}, -\frac{1}{11} - \frac{1}{11} = x(\vec{a} \cdot \vec{d}) + z$$

$$\Rightarrow \frac{-2}{11} = x(\vec{a} \cdot \vec{d}) + z$$

$$\Rightarrow x = z$$

$$\text{From equation (ii) } x = z = -5$$

$$\text{Now, } \frac{-2}{11} = -5 + (-5)(\vec{a} \cdot \vec{d})$$

$$\frac{-2}{11} + 5 = (-5)(\vec{a} \cdot \vec{d})$$

$$\frac{-53}{55} = \vec{a} \cdot \vec{d}$$

$$\Rightarrow p + q = 108$$

8. Ans (7)

$$\sum_{r=2}^{\infty} \cot^{-1}(r^2 - 5r + 7)$$

$$= \sum_{r=2}^{\infty} \tan^{-1} \left(\frac{1}{r^2 - 5r + 7} \right)$$

$$= \sum_{r=2}^{\infty} \tan^{-1} \left(\frac{1}{1 + (r^2 - 5r + 6)} \right)$$

$$= \sum_{r=2}^{\infty} \tan^{-1} \left(\frac{(r-2) - (r-3)}{1 + (r-2)(r-3)} \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{r=2}^n (\tan^{-1}(r-2) - \tan^{-1}(r-3))$$

$$= \lim_{n \rightarrow \infty} (0 - \tan^{-1}(-1) + \tan^{-1}1 - \tan^{-1}(0) + \tan^{-1}2$$

$$- \tan^{-1}1 + \tan^{-1}(n-2) - \tan^{-1}(n-3))$$

$$= \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$$

9. Ans (309)

D_n = de arrangement of n persons

$$K = {}^6P_5 (D_6 + D_5) = 6!(265 + 44) = 309 \times 6!$$